To Draw the Diagram of a Given Open Circuit Comprising at Least a Battery, Resistor/Rheostat, Key, Ammeter and Voltmeter

Aim: To draw the diagram of a given open circuit comprising at least a battery, resistor/rheostat, key, ammeter and voltmeter. Mark the components that are not connected in proper order and correct the circuit and also the circuit diagram.

Apparatus: A battery eliminator (0 to 6 V), rheostat, resistance box (0 to 100 ohm), key. D.C. ammeter (0-3) A and a D.C. voltmeter (0-3) V.

Theory: An open circuit is the combination of primary components of electric circuit in a such a manner that on closing the circuit no current is drawn from the battery.

CIRCUIT DIAGRAM:

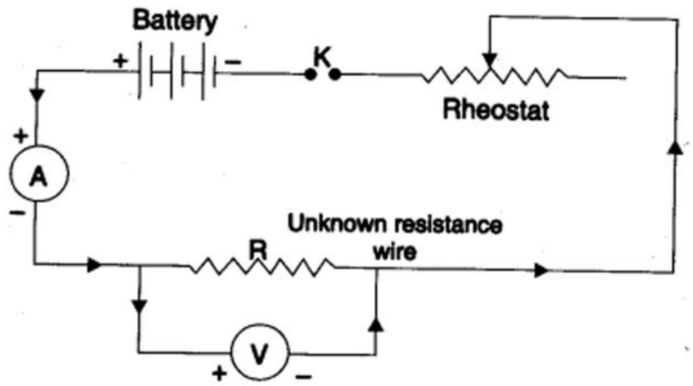
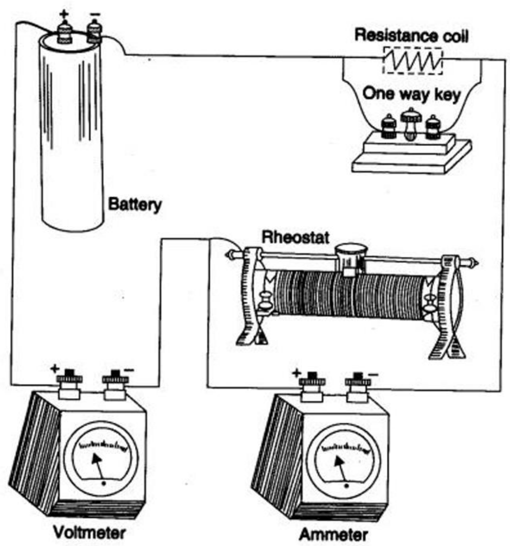


Diagram:



Open circuit diagram (Components not connected in proper order)

Procedure

1. Ammeter: It should be connected in series, with the battery eliminator.
2. Voltmeter: It should be connected in parallel to the resistor.
3. Rheostat: It should be connected in series (in place of resistance coil) with the battery eliminator.
4. Resistance coil: It should be connected in parallel (in place of rheostat).
5. One way key: It should be connected in series to the battery eliminator.
6. Correct circuit diagram: (Components connected in proper order)